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Palaeomagnetism and geochemistry of Early Palaeozoic rocks of the Barrandian (Teplá-Barrandian Unit, Bohemian Massif): palaeotectonic implications

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Abstract

The Barrandian area (the Teplá-Barrandian unit, Bohemian Massif) provided palaeomagnetic results on Early Palaeozoic rocks and chemical data on siliciclastic sediments of both Middle Cambrian and Early Ordovician to Middle Devonian sedimentary sequences; an outcoming interpretation defined source areas of clastic material and palaeotectonic settings of the siliciclastic rock deposition.

The siliciclastic rocks of the earliest Palaeozoic sedimentation cycle, deposited in the Cambrian Příbram-Jince Basin of the Barrandian, were derived from an early Cadomian volcanic island arc developed on Neoproterozoic oceanic lithosphere and accreted to a Cadomian active margin of northwestern Gondwana.

Inversion of relief terminated the Cambrian sedimentation, and a successory Prague Basin subsided nearby since Tremadocian. Source area of the Ordovician and Early Silurian shallow-marine siliciclastic sediments corresponded to progressively dissected crust of continental arc/active continental margin type of Cadomian age. Since Late Ordovician onwards both synsedimentary within-plate basic volcanics and older sediments had been contributing in recognizable proportions to the siliciclastic rocks. The siliciclastic sedimentation was replaced by deposition of carbonate rocks throughout late Early Silurian to Early Devonian period of withdrawal of the Cadomian clastic material source. Above the carbonates an early Givetian flysch-like siliciclastic suite completed sedimentation in the Barrandian.

In times between Middle Cambrian and Early/Middle Devonian boundary interval an extensional tectonic setting prevailed in the Teplá-Barrandian unit. The extensional regime was related to Early Palaeozoic large-scale fragmentation of the Cadomian belt of northwestern Gondwana and origin of Armorican microcontinent assemblage. The Teplá-Barrandian unit was also engaged in a peri-equatorially oriented drift of Armorican microcontinent assemblage throughout the Early Palaeozoic: respective palaeolatitudes of 58°S (Middle Cambrian) and 17°S (Middle Devonian) were inferred for the Barrandian rocks. The Middle Devonian flysch-like siliciclastics of the Prague Basin suggest a reappearance of the deeply dissected Cadomian source area in a proximity of the Barrandian due to early Variscan convergences and collisions of the Armorican microcontinents. Significant palaeotectonic rotations are palaeomagnetically evidenced to take place during oblique convergence and final docking of the Teplá-Barrandian microplate within the Variscan terrane mosaic of the Bohemian Massif.
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1. Introduction

Early Palaeozoic sedimentary and volcano-sedimentary sequences of the Barrandian transgressively overlie Neoproterozoic basement of the Teplá-Barrandian unit (the north-western part of the Bohemian Massif interior); the Neoproterozoic/Cambrian boundary is marked by an angular unconformity (e.g., Chlupáč et al., 1998). The Neoproterozoic basement experienced simple folding and very low-grade metamorphism during Cadomian orogeny, and was not affected by Variscan HT-LP metamorphism in general (Cháb and Suk, 1978; Cháb, 1979). However, the north-western, south-western and south-eastern margins of the Teplá-Barrandian unit, juxtaposed with the surrounding Saxothuringian and Moldanubian high-grade units, were significantly modified throughout the Variscan metamorphism and deformation. The Early Palaeozoic sequences of the

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Barrandian were left unmetamorphosed and rather weakly deformed by the Variscan cycle (e.g., Chlupáč et al., 1998; Kachlík and Patočka, 1998).

The Teplá-Barrandian unit is one of the easternmost relics of the Avalonian-Cadomian tectono-stratigraphic belt amalgamated at ≈670–550 Ma and later dismembered between north-eastern North America and western to central Europe. The Cadomian relics formerly occupied positions adjacent to the West African craton of northwestern Gondwana (e.g., Nance and Murphy, 1994; Mallard and Rogers, 1997). In the earliest Palaeozoic a large-scale rifting and fragmentation of the Cadomian belt of the northwestern margin of Gondwana produced a number of microplates involving the "Armorican Archipelago" (Franke et al., 1999) or Armorican assemblage of microcontinents (terranes) distinguished by presence of common Cadomian basements (cf., Pharaoh, 1999; Winchester et al., 2002). The peri-Gondwanan microcontinents were incorporated into Variscan belt as a chain of Cadomian terranes distributed in W–E direction from the Armorican to Bohemian Massifs (e.g., Havlíček, 1976; Cháb, 1979; Krs, 1986; Edel and Weber, 1995; Kříbek et al., 2000).

The purpose of this paper is to compare palaeomagnetic results obtained on the Barrandian Lower Palaeozoic (Tables 1–3) and geochemical data on siliciclastic rocks of both Middle Cambrian and Early Ordovician to Middle Devonian sedimentary sequences of the Barrandian (Table 4) to constrain source areas of the sedimentary material and palaeotectonic regimes of depositional environments as well as significance of the constrains for pre-Variscan and Variscan evolution of the Armorican assemblage of microplates.

2. Geological setting of the Tepl-Barrandian Unit

The Early Palaeozoic sequences of the Barrandian are of the Cambrian and Early Ordovician to Middle Devonian ages, respectively, and attain up to several km in total thickness (Chlupáč et al., 1998; Fig. 1). Principal sedimentation area of the earliest Palaeozoic sedimentation cycle, deposited under both marine and nonmarine settings, was the Cambrian Příbram-Jince Basin. At the end of Cambrian, the sedimentation was terminated, and an inversion of relief evolved during the Bohemian phase. The Příbram-Jince area became an elevation; since Tremadocian a longitudinal successory depression of extensional Prague Basin began to subside in axial parts of the Barrandian (Havlíček, 1981).

The Prague Basin axis is about 15° rotated relative to the axis of the Cambrian troughs. Shelfal siliciclastic sediments were accumulated there from Tremadocian to Early Silurian. Ordovician to Silurian evolution of geochemical features of essentially within-plate basic volcanic rocks of the Barrandian as well as

Table 1
Palaeomagnetic pole positions of Carboniferous rocks from the Bohemian Massif

	Reference no.	Age, lithology	Geographic co-ordinates		Mean palaeomagnetic directions		α_{95} (°)	k	n	Palaeomagnetic pole positions		Ovals of confidence		Reference
			φ (°)N	λ (°)E	D (°)	I (°)				φ_P (°)	λ_P (°)	d_m (°)	d_p (°)	
Plzeň Basin	8	Westphalian C-Stephanian, red beds	49.82	13.33	214.65	2.05	5.09	12.95	65	31.13 N	151.72 E	5.09	2.55	Krs (1968)
Kladno-Rakovník Basin	9	Westphalian C-Stephanian, red beds	50.17	14	211.72	−5.41	2.59	42.02	73	35.49 N	153.83 E	2.6	1.3	Krs (1968)
Blanice graben, northern part	10	Stephanian, red beds	50.03	14.82	210.25	−14.05	3.06	90.73	25	40.22 N	153.92 E	3.13	1.6	Krs (1968)
Boskovice graben, southern part	11	Stephanian, red beds	49.15	16.35	200.92	−4.73	3.94	33.97	40	39.92 N	168.63 E	3.95	1.98	Krs (1968)
Krkonoše Piedmont Basin	12	Stephanian, red beds	50.5	15.38	195.1	−13.66	3.42	31.22	57	44.65 N	174.07 E	3.49	1.78	Krs (1968)
Inner-Sudetic Basin	13	Westphalian B-Stephanian, red beds, tuffs	50.58	16.08	196.19	2.2	4.54	12.01	88	36.50 N	175.79 E	4.54	2.27	Krs (1968)

n—number of strata.

Table 2
Palaeomagnetic pole positions of pre-Variscan rocks from the Bohemian Massif (Barrandian, Moravo-Silesian Zone)

Region	Reference no.	Age, lithology	Geographic co-ordinates		Mean palaeomagnetic directions		α_{95} (°)	k	n, N	Palaeomagnetic pole positions		Ovals of confidence		Reference
			φ (°)N	λ (°)E	D (°)	I (°)				φ_P (°)	λ_P (°)	d_m (°)	d_p (°)	
Moravian Zone, Křtiny Quarry	16	Famennian, limestone	49.143	16.737	133.9	−19.1	5.8	25.9	$n=25$	35.17 S	77.03 E	3.15	6.05	Krs et al. (1995)
Moravian Zone, Josefov	17	Givetian, limestone	49.142	16.691	111.4	−24.4	6	11.7	$n=52$	23.58 S	98.89 E	3.44	6.43	Krs et al. (1995)
Moravian Zone, Čelechovice	18	Eifelian, limestone	49.532	17.087	104.8	−31.1	4	39.2	$n=34$	22.24 S	107.17 E	2.50	4.47	Krs et al. (1995)
Prague Basin	19	Early Devonian, mainly Pragian, limestone	50.005	14.299	203.6	−32.1	4.73	19.5	$n=49$	52.30 S	24.36 W	5.33	3	Chlupáč et al. (1990)
					202.2	−31.5	11.1	37.4	$N=6$	52.50 S	22.11 W	12.45	6.98	
Prague Basin	21	Early Ordovician, Tremadoc, Arenig, silicite, basalt tuff	49.75	13.63	131.84	−47.57	3.65	16.04	$n=99$	48.11 S	91.83 E	4.76	3.09	Krs et al. (1986)
Křivoklát–Rokycany Complex	23	Late Cambrian, andesite, rhyolite, dacite	49.978	13.81	119.6	−53.2	15	11.3	$N=10$	43.60 S	107.20 E	20.82	14.45	Krs et al. (1992)
Skryje-Týřovice Příbram-Jince areas	27	Middle Cambrian, Jince Fm., greywacke	49.943	13.747	54.37	−64.34	3.47	12.56	$n=143$	16.99 S	157.67 E	5.55	4.44	Krs et al. (1987)
Buchava-Slap locality	29	Middle Cambrian, Jince Fm., greywacke	49.943	13.747	64.83	−73.34	2.57	66.65	$n=47$	31.08 S	160.88 E	4.6	4.12	Krs et al. (1988)
Brdy Mts., Příbram syncline, locality "Kočka"	31	Early Cambrian, Paseky Shale, shale	49.617	13.786	53.2	72.2	2.2	107.7	$n=41$	58.28 N	69.17 E	3.89	3.44	Krs et al. (2001)

n—number of strata; N—number of sites; mean values of palaeomagnetic data from localities Nos. 16–19, 21, 23 and 27 are not involved in the sampling-site locations map in Fig. 1a.

Table 3
Virtual pole positions derived for the Devonian, Silurian and Ordovician rocks from the Barrandian, Bohemian Massif

Region	Reference no.	Age, lithology	Geographic co-ordinates		Mean palaeo-magnetic directions		α_{95} (°)	k	n	Palaeomagnetic pole positions		Ovals of confidence		Reference
			φ (°)N	λ (°)E	D (°)	I (°)				φ_P (°)	λ_P (°)	d_m (°)	d_p (°)	
U dubu Sedmi bratří	D1	Givetian, black limestone	49.89	14.11	305.50	26.70	14.40	9.20	19	33.3 S	85.0 E	8.50	15.60	Krs et al. (2001)
Hostim	D2	Middle Eifelian, limestone	49.90	14.10	210.60	−26.70	8.40	11.10	29	46.4 S	328.4 E	5.00	9.10	Krs et al. (2001)
Quarry "Prastav"	D3	Late Emsian–early Eifelian, limestone	49.95	14.20	195.80	−28.70	9.20	37.10	8	53.1 S	348.3 E	5.60	10.10	Krs et al. (2001)
Branická skála	D4	Early Emsian, limestone	49.91	14.10	328.50	27.30	15.00	21.00	6	46.3 S	61.2 E	8.90	16.30	Krs et al. (2001)
Karlštejn. Quarry Kosov	36V	Middle Silurian, basalt sill and contact aureole	49.95	14.20	205.70	−28.50	4.20	154.50	9	49.5 S	25.9 W	4.60	2.50	This study
Hlásná Třebáň	37V	Middle Silurian, contact aureole of basalt sill	49.93	14.22	172.80	−21.60	7.10	42.60	11	50.8 S	25.4 E	7.50	3.90	This study
Žebrák–Točník	38V	Llanvirn, pillow lava	49.88	13.90	139.70	60.20	26.30	7.40	6	60.8 S	102.1 E	39.90	30.20	This study
Točník	43V	Arenig, Klabava Fm., mineralized layers	49.88	13.90	41.90	63.70	11.30	9.70	19	11.9 N	14.8 W	17.90	14.20	This study
Ejpovice	46V	Early Arenig, Klabava Fm., tuffite	49.75	13.57	145.60	−57.80	6.50	15.00	35	61.3 S	99.9 E	9.50	7.00	This study
Komárov-Jivina	47V	Tremadoc, red greywacke	49.78	13.82	176.00	−58.40	3.50	60.10	29	78.9 S	30.2 E	5.20	3.80	This study
Žebrák–Točník	48V	Llanvirn, silicified lava	49.88	13.90	188.10	−46.80	7.80	35.90	12	67.3 S	4.9 W	10.10	6.50	This study
Praha-Libeň, Černá skála	49 V	Early Caradoc, Letná Fm., black shale	50.0	14.45	194.50	−45.80	8.10	7.10	24	64.6 S	16.9 W	10.30	6.60	This study

n—number of strata.

Table 4
Major and trace element compositions of representative samples of siliciclastic sediments of the Barrandian

Sample	SB-7	SB-6	SB-5	SB-2D	SB-10	SB-11A	SB-8
SiO_2 [wt.%]	70.98	80.28	83.07	77.52	81.33	77.45	70.82
TiO_2	0.78	0.57	0.57	0.44	0.62	0.42	0.71
Al_2O_3	9.72	7.76	9.85	7.43	8.72	8.12	10.97
FeO^t	7.67	3.53	0.83	5.98	3.30	1.07	4.62
MnO	0.050	0.036	0.025	0.262	0.028	0.003	0.088
MgO	1.23	0.76	0.10	0.96	0.62	0.57	0.84
CaO	1.45	0.19	0.09	0.57	0.22	0.09	3.38
Na_2O	1.87	0.08	0.01	0.61	1.17	0.29	0.88
K_2O	1.08	2.06	2.06	1.44	2.05	2.29	1.60
P_2O_5	0.10	0.11	0.11	0.21	0.08	0.09	0.11
CO_2	0.82	2.28	0.11	1.97	0.02	4.27	2.48
H_2O	3.13	2.04	2.02	2.63	1.97	5.25	2.70
Total	98.88	99.70	98.85	100.02	100.13	99.66	99.20
Sr [ppm]	111	313	313	97	63	76	102
Rb	129	76	76	125	134	41	57
Ba	1223	413	413	709	1154	131	297
Th	4.35	10.86	10.86	7.17	9.1	4.83	8.44
Ta	0.48	0.35	0.35	0.47	0.64	0.56	0.36
Nb	8	12	12	24	8	8	8
Zr	148	260	260	183	138	149	266
Hf	4.32	6.52	6.52	8.61	10.51	2.35	7.45
Sc	14.6	10	10.0	8.0	5.5	9.5	11.4
Cr	8	24	24	97	5	81	65
La	21.59	20.98	37.39	26.57	26.84	28.04	31.64
Ce	43.34	43.15	75.50	53.63	50.85	48.00	57.02
Nd	24.16	23.11	35.28	27.00	23.44	24.85	29.47
Sm	4.78	4.40	6.93	5.17	4.78	5.31	5.82
Eu	1.23	1.07	1.61	1.22	0.97	1.33	1.28
Gd	4.59	3.72	5.89	4.04		5.19	6.43
Tb	0.69	0.56	0.96	0.65	0.55	0.71	0.78
Yb	2.20	1.44	1.94	2.08	1.97	2.52	2.62
Lu	0.33	0.22	0.29	0.33	0.31	0.37	0.41
Y	38	22	22	32	37	16	27
Ce_N/Yb_N	4.30	6.54	8.49	5.63	5.63	4.16	4.75
Eu/Eu^*	0.78	0.78	0.74	0.78	0.65	0.75	0.63

The samples are arranged according to stratigraphic order. SB-7—subgreywacke sandstone, Middle Cambrian, Jince (corresponds to the Middle Cambrian sampling sites clustered under No. 27 in Table 2); SB-6—subgreywacke sandstone, late Llanvirn, Ejpovice (dtto sampling site 46V in Table 3); SB-5—silty shale, early Caradoc, Libeň (dtto sample site 49 V in Table 3); SB-2D—diamictite, late Ashgill (base), Levín; SB-10—quartzitic siltstone, late Ashgill, Hlásná Třebáň; SB-11A—black micaceous shale, early Llandovery, Hlásná Třebáň; (dtto sampling site 37V in Table 3); SB-8—siltstone, early Givetian, Hostim.

Silurian sedimentary record (indicating withdrawal of clastic material source regions) suggests a maximal extension of the Prague Basin throughout the Silurian period (Patočka et al., 1993; Patočka et al., 1994). At the very beginning of the Silurian, the shallow-water siliciclastic sedimentation was replaced by relatively deep-water black shales; subsequently, the black shales were substituted by carbonate suite since the late Early Silurian. Deposition of the limestones eventually gained prevalence in the Late Silurian and Early Devonian. Rapid accumulation of continental flysch-like siliciclastic sediments above the late Early Silurian to Early Devonian carbonate dominated suite in the Early Givetian heralded impingement phase of Variscan orogeny (Kukal and Jäger, 1988); in response, the extension, sedimentation and volcanic activity of the Prague Basin ceased.

3. Outline of palaeomagnetism on the Bohemian Massif

According to palaeomagnetic data the Bohemian Massif is a part of stable European lithospheric plate since Early Permian. The tectonically stable European plate is well documented by palaeomagnetism since the same period (e.g., Krs, 1968; Van der Voo, 1990; Torsvik et al., 1990, 1992; Krs et al., 1997). However, the palaeomagnetic results obtained from pre-Early Permian Palaeozoic rocks of the Bohemian Massif show continuous changes of palaeoinclinations as well as large-scale palaeorotations. For example, an influence of palaeotectonic rotations on the scatter of Late Carboniferous palaeomagnetic data from the Bohemian Massif may be demonstrated on values in Table 1: an azimuthal difference between palaeomeridians is up to 17° and a difference between palaeolatitudes is 9°.

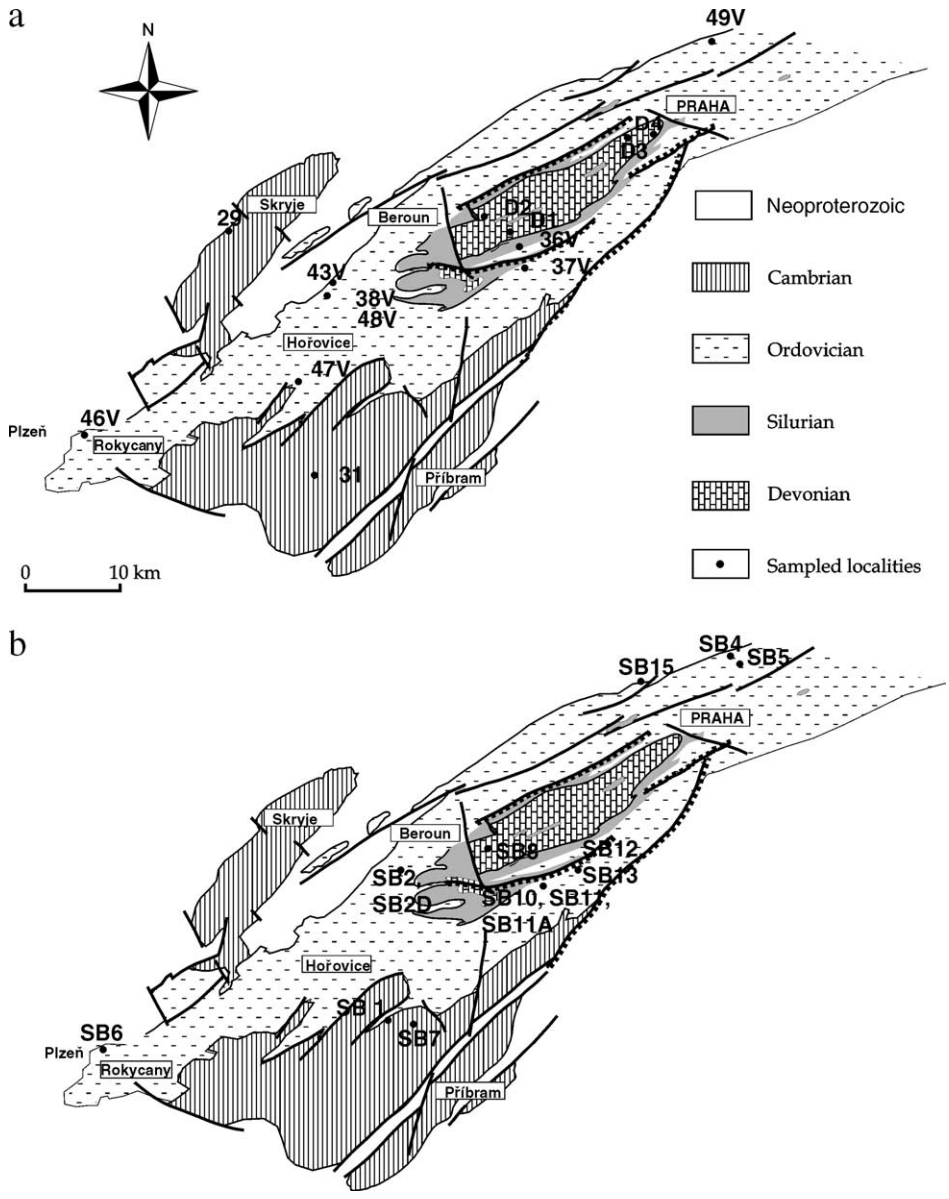


Fig. 1. The generalized geological maps of the Barrandian Lower Palaeozoic showing the palaeomagnetic sampling sites (a) (for sampling site names and further details see Tables 2 and 3), and locations of the samples collected for chemical analyses (b) (for locality names and further details see Table 4).

In the west European Variscides, the palaeomagnatic data usually show prominent differences in local palaeomeridian orientations for periods older than the Early Permian as a consequence of horizontal palaeotectonic rotations. The results on Middle Carboniferous rocks yielded mean values of clockwise rotation of ≈50°, and the data on Early Carboniferous ones provided extreme values of 120° (Edel, 1987). In the Bohemian Massif similar and even higher values of anomalous palaeomagnetic declinations of 210° (i.e., the anomalous palaeorotations) were reported for Early Ordovician and Devonian rocks from the Barrandian area (Chlupáč and Krs, 1967; Krs et al., 1986; Mauritsch and Pruner, 1990). Palaeomagnetic data on the Barrandian sequences may indicate an anomalously high palaeolatitudinal drift, from 58°S for Early Cambrian to 17°S for Middle Devonian (Krs and Pruner, 1995; Krs et al., 2001).

4. Palaeomagnetic methods

Remanent magnetization and volume magnetic susceptibility were measured using JR-5 high sensitivity spinner magnetometers and a KLY-2 kappa-bridge, respectively (Jelínek, 1966, 1973). The MAVACS (Magnetic Vacuum Control System) apparatus was used for progressive thermal demagnetization (Příhoda et al.,

1989). Zijderveld plots of vector of remanent magnetization were constructed for all specimens investigated, and multi-component analysis of remanence was performed after Kirschvink (1980). The statistically calculated mean directions (Fisher, 1953) of separated remanence components were evaluated in combination with fold tests.

Horizontal palaeotectonic rotations of rock formations of different ages are observed in the Barrandian area. The A-component of remanence, originated in Recent, was interpreted in the temperature interval of 20–120 °C, the B-component, corresponding to late Variscan overprint, was interpreted in the low temperature interval (LTC) and the C-component, corresponding to palaeomagnetization, was separated in the high interval of temperature (HTC).

5. Palaeogeographic interpretation of palaeomagnetic results on Early Palaeozoic rocks of the Tepl-Barrandian Unit

Palaeomagnetic declination provides evidence of angular deviations resulting from horizontal palaeotectonic rotations of a particular rock formation from time of origin to present, including rotations of lithospheric plate (or terrane) where the individual formation was tectonically incorporated. According that, the palaeomagnetic declinations obtained on the Bohemian Massif rocks of pre-Early Permian ages reflect not only the azimuth of palaeomeridians for the European Plate but also a sum of all horizontal palaeotectonic rotations of the respective rock formations which occurred before the end of Variscan orogeny.

The Variscan amalgamation of the Bohemian Massif terrane mosaic during the stable European plate origin (see above) produced a number of palaeotectonic rotations occurring throughout interactions among the accreting terranes, including the Teplá-Barrandian Unit as a carrier of the Early Palaeozoic sequences of the Barrandian (Fig. 2). Predominantly clockwise rotations were palaeomagnetically established for the investigated rocks of the Barrandian area (as well as for the other Variscan and pre-Variscan formations of the Bohemian Massif) (e.g., Krs and Pruner, 1995; Krs et al., 2001). With respect to the deformational effects of the Variscan orogeny on the Barrandian rocks and to palaeogeographic affinities of the studied Early Palaeozoic formations, the Variscan rotations cannot be commuted with, for example, Ordovician and Silurian rotations of either the Teplá-Barrandian Unit or even the whole Bohemian Massif (cf. Tait et al., 1994a,b, 1995). Palaeotectonic rotations of a particular rock formation can be documented only on condition that palaeomagnetic directions are investigated separately.

5.1. Middle to Late Cambrian

The Late Cambrian volcanics of the Křivoklát–Rokycany Complex (mostly andesites, in minor part rhyolites and dacites) show different values of palaeomagnetic directions for different localities, and are an example of the palaeotectonic rotations in the Barrandian (Krs et al., 1987, 1988). Palaeomagnetic declination inferred from

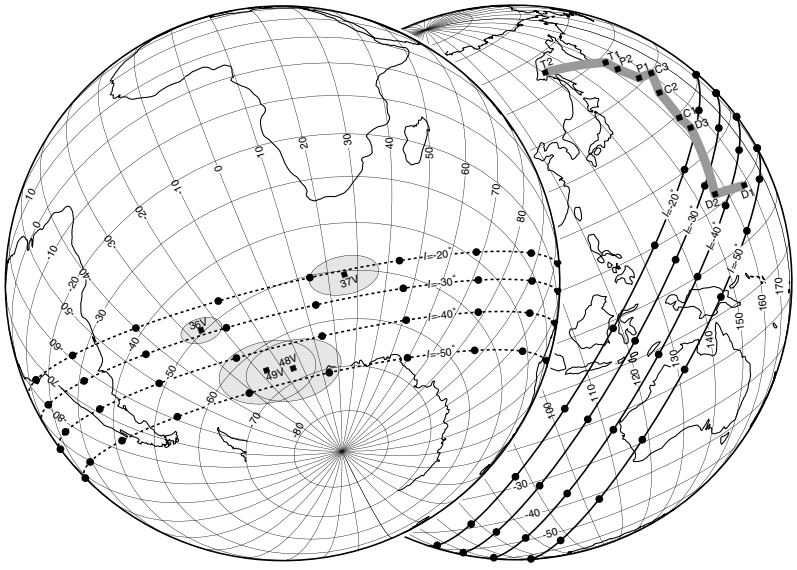


Fig. 2. Middle to Late Ordovician and middle Silurian virtual pole positions, the Barrandian area, Bohemian Massif: 36V—Karlštejn, middle Silurian, contact aureole of basalt sill; 37V—Hlásná Třebáň, middle Silurian, contact aureole of basalt sill; 48 V–Žebrák—Točník, Llanvirn, silicified basalt lava; 49 V—Prague, "Černá skála", Early Caradoc, black shale. Apparent Polar Wandering Path inferred from East European Craton for Early Devonian to Middle Triassic time span is presented by a thick grey line. Full small circles denote pole positions for palaeomagnetic declination step of $\Delta D = 20°$. Dotted lines indicate distributions of pole positions due to palaeotectonic rotations for the same palaeomagnetic inclination.

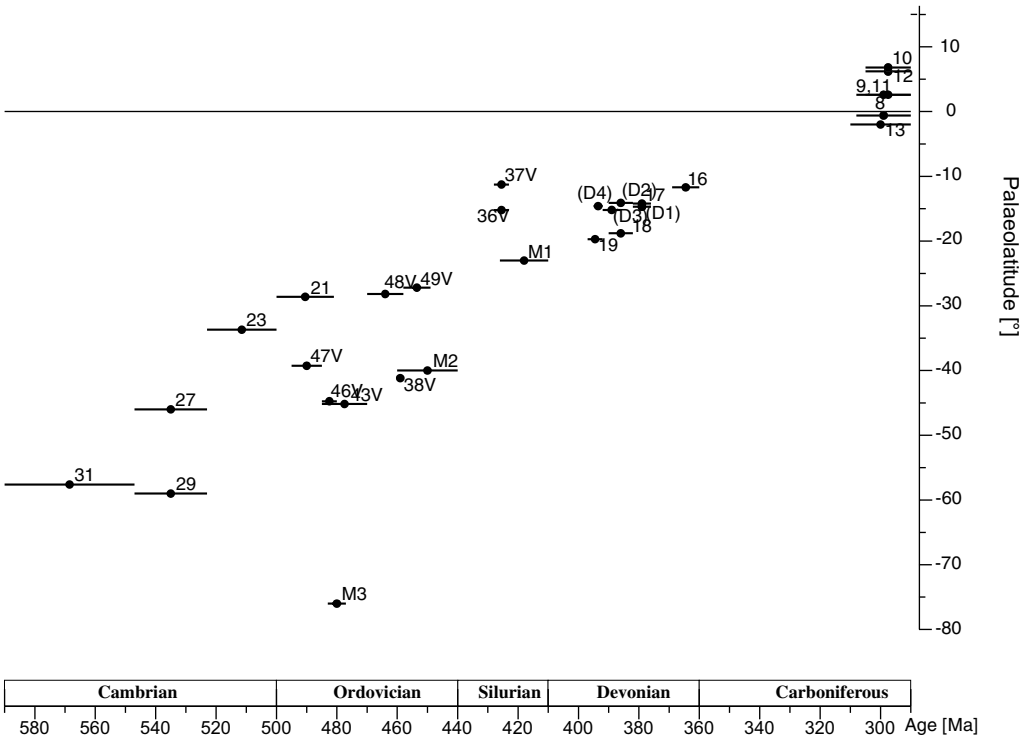


Fig. 3. The Barrandian area, Bohemian Massif: palaeolatitudes extrapolated from different sources, recalculated to a single reference point at $\varphi = 50°N$, $\lambda = 14°E$. Data were recently inferred from virtual pole positions for the Ordovician and middle Silurian rocks (36 V, 37 V, 48 V and 49 V). Full circles belong to values derived from databases (Krs and Pruner, 1995; Krs et al., 2001). Values M1, M2 and M3 follow Tait et al. (1994a,b, 1995). All data were recalculated according to geochronological scale given in the Geological Time Table published by Haq and Van Eysinga (1998). (The Late Cambrian and Ordovician rocks should be subjected to additional palaeomagnetic investigations because the palaeolatitudes under reference Nos. 21 and M3 were inferred from palaeomagnetic directions of only single polarity).

mean value of the palaeomagnetic directions (calculated for all localities) provides none information on differential values of declination or palaeomagnetic rotations among the individual localities (Table 2). Scatters of virtual pole positions (Krs et al., 1992) are generally larger than those of palaeomagnetic pole positions. In spite of that, differences among the palaeomagnetic declinations (resulting from the palaeotectonic rotations) mostly extend beyond the scatters of virtual pole positions. The data indicate that the Křivoklát–Rokycany Complex did not rotate horizontally as a uniform block. Palaeolatitudes (58°S—see references Nos. 27 and 29, and reference No. 23, respectively, Table 2 and Fig. 3) inferred for the Middle to Late Cambrian sedimentary and volcanic rocks of the Barrandian area (showing normal and reverse palaeomagnetic polarities) are in agreement with palaeolatitudes recalculated for the Gondwana-derived Armorican assemblage of microplates and the Gondwana supercontinent (e.g., Van der Voo, 1993; Franke et al., 1999; Pharaoh, 1999).

5.2. Middle to Late Ordovician

Low values (29°S) of the Middle to Late Ordovician palaeolatitudes resulting from palaeomagnetic directions of both normal and reverse polarity of original remanence components were obtained on silicities and shales of the "Žebrák–Točník" and "Černá skála" localities, respectively. The results are not compatible with new data gained on pillow lava sample of the "Žebrák–Točník" locality (No. 38 V in Fig. 3 and Table 3) as well as with the Late Ordovician palaeolatitude (marked by symbol M2 in Fig. 3) after Tait et al. (1994a). The LTC-component was based on demagnetization of rocks from the localities "Točník" (No. 43 V), "Ejpovice" (No. 46 V), "Žebrák–Točník" (No. 48 V) and "Praha-Libeň, Černá skála" (No. 49 V) in temperature interval of 80 (160 °C) to 250 °C. The HTC-component was obtained on rocks from the localities "Žebrák–Točník" (No. 48 V) and "Točník" (No. 43 V) and from "Praha-Libeň, Černá skála" (No. 49 V) and "Ejpovice" (No. 46 V) (Table 3); the temperature interval of the former demagnetization was 350 to (620 °C) 640 °C, and of the latter 300 to (480 °C) 540 °C. Significantly low palaeolatitude given under reference No. 21 in Fig. 3 evidently results from subsequent remagnetization of the Early Ordovician silicites (Krs et al., 1986; Mauritsch and Pruner, 1990). On the other hand, peri-polar palaeolatitude values of 76°S for the Early Ordovician rocks (marked by symbol M3 in Fig. 3) according to Tait et al. (1994b) are in agreement

with palaeoclimatic indicators provided by the Early Ordovician sediments but imply an anomalously high drift for this period.

5.3. Middle Silurian

Palaeolatitudes of 14°S, calculated for the middle Silurian rocks of the Barrandian (samples collected from basalt sill and surrounding contact aureole, localities "Karlštejn", No. 36 V, and "Hlásná Třebáň", No. 37 V—Table 3), are lower than palaeolatitudes of ca. 17°S based on data measured on rocks of Devonian age (Fig. 3). The LTC-component was based on demagnetization of rocks from the localities "Karlštejn" (No. 36 V) and "Hlásná Třebáň" (No. 37 V) in temperature interval of 60 (90 °C) to 200 °C (300 °C) and the HTC-component was gained in temperature interval of 400 (440 °C) to (540 °C) 580 °C. Scatters of virtual pole positions are generally smaller than those of palaeomagnetic pole positions for Devonian. An influence of palaeotectonics on scatter of the middle Silurian palaeomagnetic data is presented in Table 3. An azimuthal difference between palaeomeridians is up to 33° whereas the difference between palaeolatitudes is only 5°.

5.4. Early to Middle Devonian

An analysis of natural remanent magnetization of the Early to Middle Devonian limestones of the Barrandian area is based on a set of samples from several localities (Tables 2 and 3). B-component was separated from the limestones in temperature interval of 120 °C (160 °C) to 420 °C and corresponds to the Variscan overprint (see Table 8, p. 35 in Krs and Pruner, 1995). C-component was separated in temperature interval of 420–550 °C and corresponds to the Early or Middle Devonian palaeomagnetic directions. Palaeomagnetic pole positions calculated from mean values of the palaeomagnetic directions of the C-components of remanence (corrected for the dip of strata) clearly indicate prominent horizontal palaeotectonic rotations during the Variscan orogeny. The rotations vary in magnitude from one locality to another; their values are high for the Devonian limestone formations of the Moravo-Silesian Zone and reach extreme values (difference between palaeomaridians is 63°) on three localities out of the total of five studied in the Barrandian area (Tables 2 and 3). Peri-equatorial palaeolatitude of 17°S was derived from data obtained from the Devonian rocks (Krs and Pruner, 1995; Krs et al., 2001).

6. Chemical methods

Fourteen samples of the Early Palaeozoic siliciclastic sediments were collected in the Barrandian. Major element concentrations were determined in the Laboratory of Chemical Methods, Insitute of Rock Structure and Mechanics, AS CR (head of the laboratory Ing. V. Chalupský) using classical wet chemical analysis. Trace elements were analyzed by X-ray fluorescence (Sr, Ba, Nb, Zr, Y and Cr) in the Gematest Ltd., Prague (by Ing. J. Štrublová), and by INAA (Rb, Th, Ta, Hf, Sc and REE) in the Laboratory of Nuclear Spectroscopy of the Institute of Nuclear Physics, AS CR, Prague. Details and precision of the method are described in Vobecký et al. (1971) and Kuncíř et al. (1980). Examples of analytical results are shown in Table 4.

7. Geochemistry of Early Palaeozoic siliciclastic sediments of the Tepl-Barrandian Unit

Concentrations of some less mobile trace elements (of HFS and REE groups and some mantle compatible elements) are considered to indicate genetic links of accumulations of sedimentary rocks to provenances and paleotectonic regimes (e.g., Bhatia, 1985; Bhatia and Crook, 1986; Floyd and Leveridge, 1987; Floyd et al., 1991; McLennan et al., 1993; McLennan, 2001).

Based on concentrations and ratios of the less mobile trace elements as well as on stratigraphic positions, a succession of five groups of Early Palaeozoic siliciclastic sediments was distinguished in the Barrandian area. Age values corresponding to individual periods and intervals (given in brackets) are after Rename et al. (2000).

7.1. Middle Cambrian (≈515 Ma)

Siliciclastic sediments of the Middle Cambrian age (sample SB-7 in Table 4) are generally depleted in most of LIL elements as well as in Ta and Nb relative to the average shale composition (Bowen, 1979); in contents of the other HFSE and also in concentrations of mantle compatible elements the siliciclastics are similar to the latter standard (Fig. 4a). They often show REE concentrations lower than the post-Archaean Average Australian Sedimentary rock (PAAS) (Nance and Taylor, 1976) (Fig. 5a), and very low values of the chondrite-normalized (Anders and Grevesse, 1989) Ce_N/Yb_N ratios. Compared to the upper continental crust (Taylor and McLennan, 1985; McLennan, 2001), the siliciclastic rocks display less fractionated lithophile elements relative to mantle compatible ones—as demonstrated, for example, by low Th/Sc, La/Sc, Nb/Y, Hf/Y and Hf/Ti values. The Middle Cambrian siliciclastics were presumably derived from primitive crust of island-arc type according to features presented above (cf. Bhatia, 1985; Bhatia and Crook, 1986; Floyd et al., 1991; McLennan et al., 1993) (Fig. 6).

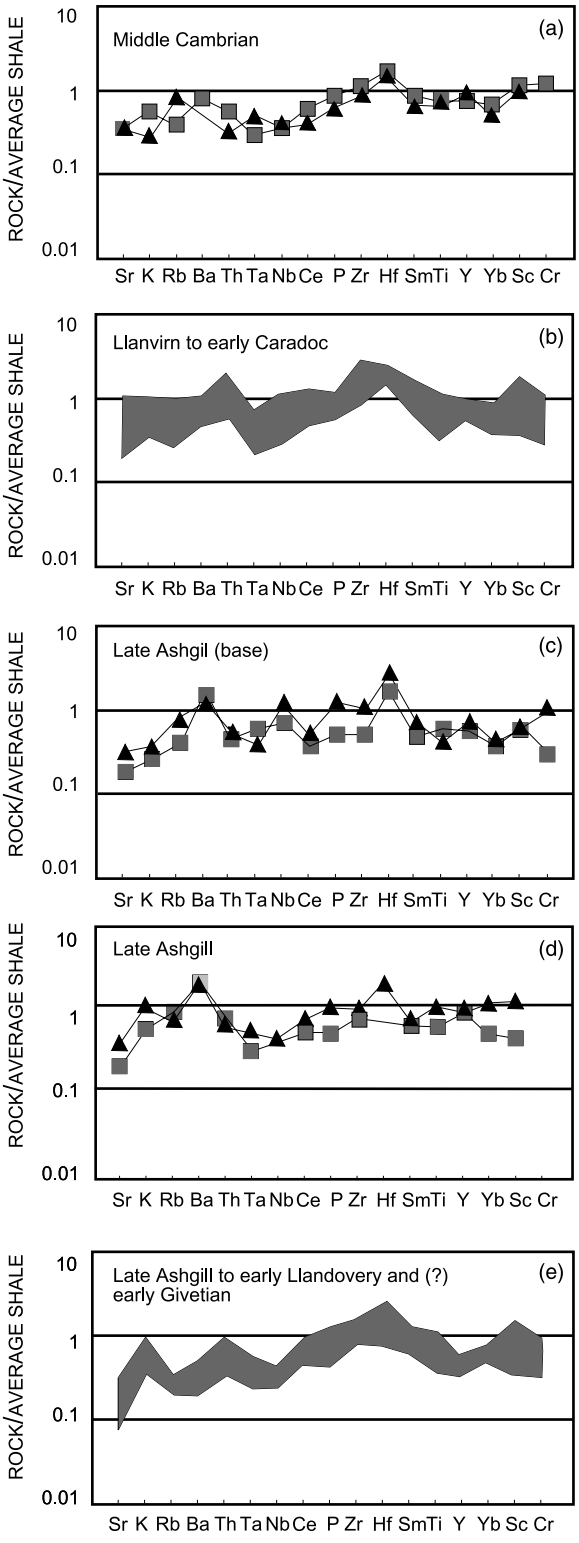


Fig. 4. Distribution patterns of trace element concentrations in the Early Palaeozoic siliciclastic sediments of the Barrandian normalized by the average shale values after Bowen (1979).

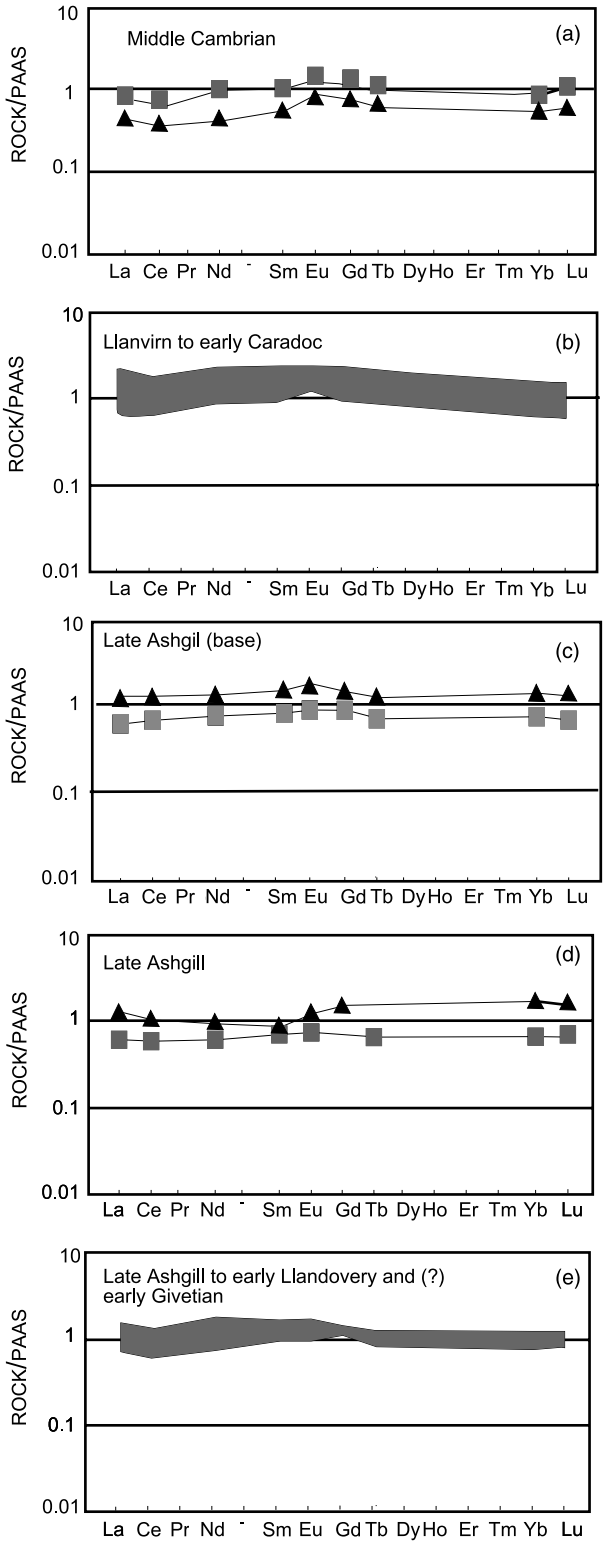


Fig. 5. Distribution patterns of REE concentrations in the Early Palaeozoic siliciclastic sediments of the Barrandian normalized by the post-Archaean Average Australian Sedimentary rock (PAAS) values after Nance and Taylor (1976).

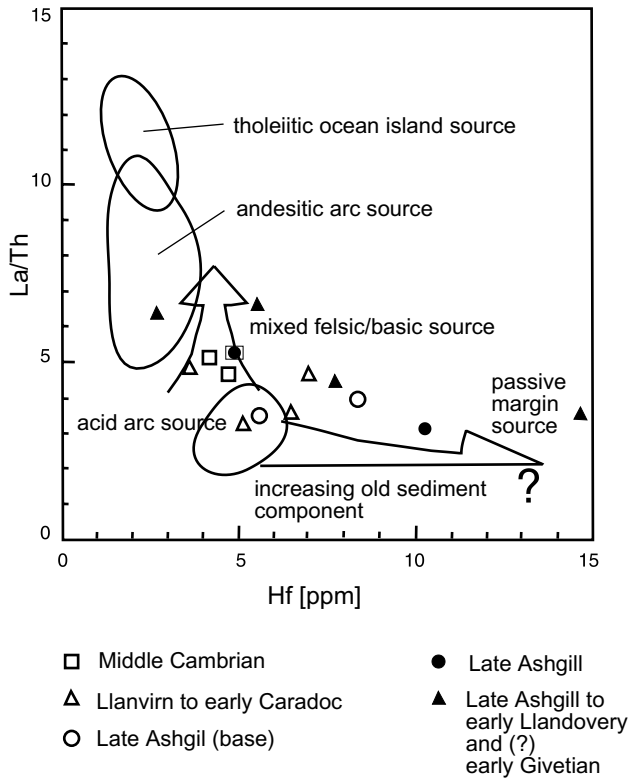


Fig. 6. The Early Palaeozoic siliciclastic sediments of the Barrandian in Hf versus La/Th diagram (cf. Floyd and Leveridge, 1987; Floyd et al., 1991).

7.2. Middle Ordovician (Llanvirn to early Caradoc) (462–453 Ma)

The siliciclastic rocks sedimented in the Middle Ordovician (samples SB-5 and -6) are slightly depleted in most of LILE (except Th), moderately depleted in Ta, Nb and some transitional elements relative to the average shale; nevertheless, some samples are very similar to the latter standard (Fig. 4b). The siliciclastic sediments show PAAS-like concentrations of lanthanides (Fig. 5b), and exhibit the highest Ce_N/Yb_N values. The provenance of the Middle Ordovician sediments was presumably an evolved crust of continental arc to active continental margin type (cf. Bhatia and Crook, 1986; Floyd et al., 1991; McLennan et al., 1993). Dacites to rhyolites of Late Cambrian volcanic centres, which were superimposed on the Neoproterozoic basement of the Teplá-Barrandian unit (e.g., Waldhausrová, 1971), substantially contributed to the clastic material and enhanced the volcanic/plutonic arc-like features of the sediments (Fig. 6).

7.3. Late Ordovician (early late Ashgill) (ca. 437 Ma)

The siliciclastic sediments of the early late Ashgill age (sample SB-2D) are mostly depleted in LILE (except Ba), Ta and Nb, and in most of mantle compatible elements compared with the average shale (Fig. 4c). In relation to REE concentrations in the PAAS, the studied rocks are either slightly depleted or nearly equivalent to the latter standard (Fig. 5c); however, they display lower values of Ce_N/Yb_N compared with the Middle Ordovician siliciclastics. The source of the early late Ashgill siliciclastics corresponded to the continental arc/active margin-like provenance of the Middle Ordovician sediments. An admixture of some older (recycled) sedimentary component was already present in some of the early late Ashgill siliciclastic rocks (cf. Floyd and Leveridge, 1987; Fig. 6). A less advanced fractionation of LREE relative to HREE as well as low Hf/Y and Nb/Y values of the siliciclastics indicate some input of eroded basic volcanic rocks of Ordovician age.

7.4. Late Ordovician (late Ashgill sensu stricto) (ca. 435 Ma)

The siliciclastic rocks deposited in the late Ashgill (sample SB-10) display lower concentrations of LILE (with exception of Ba), Ta and Nb as well as slightly depleted or nearly identical contents of transitional elements relative to the average shale (Fig. 4d). In general, they have PAAS-like REE concentrations (Fig. 5d), but some of them show low degree of fractionation of LREE vs HREE (likewise the siliciclastics of the early late Ashgill age). The concentrations of trace elements and lanthanides in the youngest Ordovician sediments suggest permanence of the progressively dissected source area of continental arc/active continental margin type (see above). As indicated by conspicuously high Th/Sc, La/Sc and La/Nb ratios as well as Hf vs. La/Th relation (Fig. 6), the older sedimentary material contributed in larger volume to this group of siliciclastics. Supply of eroded Ordovician basic volcanics resulted in high Ti/Nb, and low Nb/Y, Hf/Ti, Hf/Y and Ce_N/Yb_N ratios of the studied rocks.

7.5. The latest Ordovician (i.e., the latest Ashgill) to Early Silurian (Llandovery), and possibly to Middle Devonian (Early Givetian) (?) (434-378 Ma)

In spite of very long time span, the siliciclastic samples of this group (samples SB-8 and -11A) display nearly identical chemical characteristics. Compared with the average shale composition, the samples are depleted in LILE, Ta, Nb and mostly also in mantle compatible elements (Fig. 4e). Lanthanide contents vary around the PAAS values in these siliciclastics (Fig. 5e). The sediments also show rather low Ce_N/Yb_N ratios. The trace element and REE signatures of the studied samples evidence a steady growth of input of the older sedimentary component (see a significantly elongated distribution of the samples along Hf axis in Fig. 6) to the clastic material provided for the most part by proceeding erosion

of the old source area of continental arc/active margin type. The siliciclastic sediments show high Ti/Nb and low Nb/Y, Hf/Ti, Hf/Y and Ce_N/Yb_N ratios inherited from wide spectrum of the Barrandian basic volcanic rocks of Early Ordovician to Middle Devonian ages.

8. Discussion and conclusion

Palaeomagnetic data on Early Palaeozoic rocks of the Teplá-Barrandian unit and recalculated data on contemporaneous units of the Variscan Europe are in good agreement with regard to the palaeolatitudinal drift values in the Middle Cambrian to Middle Devonian time span. Contradictory results were found only for the Ordovician (particularly for the Early Ordovician) sequences. Peri-polar values of the Early Ordovician palaeolatitudes (76°S—Tait et al., 1994a,b, 1995) are consistent with the indicators of cold climate identified in the Early Ordovician rocks. However, they mean also an anomalously fast palaeolatitudinal drift of the Teplá-Barrandian unit (as well as of Gondwana and Armorican assemblage of microplates). Solution of this problem requires additional palaeomagnetic and palaeoclimatologic studies (moreover, the palaeolatitudes marked as 21 and M3 in Fig. 3 are based on palaeomagnetic directions with only single polarity). Palaeolatitudes for both Middle Cambrian sediments and Late Cambrian volcanics of the Barrandian were inferred from normally and reversely polarized palaeomagnetic directions supported by fold tests. The palaeolatitudes correspond to those identified for the Armorican microplates and Gondwana (cf., Van der Voo, 1993; Franke et al., 1999; Pharaoh, 1999).

The siliciclastic rocks of the Barrandian sedimentary sequences of both Middle Cambrian and Early Ordovician to Middle Devonian ages provided a sum of chemical data which helped to elucidate source areas of the sedimentary material and palaeotectonic settings of respective depositional environments (Table 4 and Figs. 4–6).

The Middle Cambrian siliciclastic rocks were sedimented during the oldest Early Palaeozoic sedimentation cycle principally in the Příbram-Jince Basin. The most probable Precambrian source area of the sediments was an early Cadomian volcanic island arc developed on Neoproterozoic oceanic lithosphere, and accreted to the Cadomian active margin (Jakeš et al., 1979; Matějka, 1981; Nance and Murphy, 1994; Zulauf et al., 1999); the volcanic-arc related subduction and accretion are dated by Dörr et al. (2002) at >550 Ma and 550–540 Ma, respectively.

Inversion of relief in the Barrandian area interrupted the sedimentation and uplifted the Cambrian basin; the successory extensional Prague Basin subsided there since Tremadocian. Shallow-marine siliciclastic sedimentation prevailed in the Ordovician. Deeper-water black-shale deposition followed in the Early Silurian, and in turn was replaced by sedimentation of carbonate rocks in the late Early Silurian (Chlupáč et al., 1998). Within-plate basic volcanic rocks of the Barrandian evolved in chemical composition from dominantly alkaline to mostly transitional and tholeiitic since the Ordovician to Silurian (Patočka et al., 1993), and together with the described Silurian sedimentary record indicate that the Prague Basin extension reached its maximum in the latter period (Patočka et al., 1994; Kachlík and Patočka, 1998).

The Early Ordovician to Early Silurian sediments of the Prague Basin in majority represent an accumulation of the siliciclastic material eroded from an older crust of continental arc/active continental margin type dominated by intermediate to acid (meta)igneous rocks. The older crust may be identified with the Cadomian-age crust of peri-Gondwanan fragments, involving the Neoproterozoic basement of the Teplá-Barrandian unit (e.g., Nance and Murphy, 1994; Edel and Weber, 1995; Keppie and Dostal, 1998; Kříbek et al., 2000). The Cadomian crust had been progressively extended and attenuated both by extension and erosion since the times of northwestern Gondwana breakdown in the earliest Palaeozoic (cf., Pharaoh, 1999). The siliciclastic sediments of the Middle Ordovician age reflect an intense contribution of clasts derived from dacitic to rhyolitic volcanic rocks of the Barrandian Upper Cambrian (cf., Waldhausrová, 1971; Vidal et al., 1975). Synsedimentary basic volcanics of the Early Ordovician to Silurian ages of extrusion (Patočka et al., 1993; Chlupáč et al., 1998) had been providing steady input into the siliciclastic rocks since the Late Ordovician onward, and a supply of recycled intrabasinal sedimentary component had been also increasing in the sediments since the same period.

Above the Late Silurian and Early Devonian limestone suite of the Prague Basin (deposited throughout a period of withdrawal of the Cadomian clastic material source) the flysch-like siliciclastics of the early Givetian age completed the Early Palaeozoic sedimentation of the Barrandian, and marked an onset of the Variscan orogeny (Kukal and Jäger, 1988; Chlupáč et al., 1998). It has to be pointed out that chemistry of the Middle Devonian siliciclastics shows many features identical to those of the latest Ordovician to Early Silurian siliciclastic rocks. Chemical similarity of the Middle Devonian and the latest Ordovician to Early Silurian siliciclastics may be interpreted as a result of emergence of already deeply dissected Cadomian crust of continental arc/active continental margin type in proximity of the Prague Basin in the mid-Devonian times (e.g., Strnad and Hladil, 2001).

The comparison of palaeomagnetic and geochemical results obtained on the Early Palaeozoic rocks of the Barrandian suggests that since the Middle Cambrian to Early/Middle Devonian boundary interval the extensional palaeotectonic regime of the Teplá-Barrandian unit (e.g., Patočka et al., 1993; Kachlík and Patočka, 1998; Chlupáč et al., 1998) was related to the Early Palaeozoic pervasive extension and fragmentation of the northwestern margin of Gondwana supercontinent and origin of the Armorican microcontinent assemblage (cf., Franke et al., 1999; Pharaoh, 1999; Winchester et al., 2002). As a component of the Armorican assemblage of microcontinents, the Teplá-Barrandian unit also experienced a substantial palaeolatitudinal drift towards the periequatorial realms: the palaeolatitudes of 58°S and 17°S were obtained for the Middle Cambrian and Middle Devonian rocks of the Barrandian, respectively (Krs, 1986; Krs and Pruner, 1995; Krs et al., 2001). On the other hand, the Middle Devonian (early Givetian) flysch-like siliciclastic sediments indicate the reappearance of the Cadomian clastic material source area in the vicinity of the Barrandian (see above). The emergence of the old landmasses may be related to early Variscan and Variscan convergences and collisions among the Armorican microplates, and resulting amalgamation of the Variscan terrane mosaic of the Bohemian Massif (e.g., Ziegler, 1989; Tait et al., 1996; Keppie and Dostal, 1998; Winchester et al., 2002). These processes involved palaeomagnetically documented large-scale palaeotectonic rotations (Tables 2 and 3) (e.g., Krs et al., 2001) interpreted as oblique convergence and final docking of the Teplá-Barrandian unit microplate and the Bruno-Silesia (Hladil et al., 1999).

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References

Anders, E., Grevesse, N., 1989. Abundances of the elements: Meteoritic and solar. Geochimica et Cosmochimica Acta 53, 197–214.

Bhatia, M.R., 1985. Composition and classification of Palaeozoic flysch mudrocks of eastern Australia: implications in provenance and tectonic setting interpretation. Sedimentary Geology 41, 249–268.

Bhatia, M.R., Crook, K.A.W., 1986. Trace element characteristics of graywackes and tectonic setting discrimination of sedimentary basins. Contributions to Mineralogy and Petrology 92, 181–193.

Bowen, H.J.M., 1979. Environmetal Chemistry of the Elements. Academic Press, London, New York, Sydney. 333 pp.

Cháb, J., 1979. Upper Proterozoic of the Bohemian Massif and its significance for evolution of the European continent. In: Collected Abstract of the Workshop on the 60th Anniversary of the Geological Survey, Prague, pp. 23–29.

Cháb, J., Suk, M., 1978. The metamorphic development of the Bohemian Massif on the Czechoslovak territory. Sborník geologických Věd, Geologie 31, 109–126.

Chlupáč, I., Havlíček, V., Kříž, J., Kukal, Z., Štorch, P., 1998. Palaeozoic of the Barrandian (Cambrian–Devonian). Geological Survey, Prague. 183 pp.

Chlupáč, I., Krs, M., 1967. Paläomagnetismus und Paläoklimatologie des Devons. Geologie 16, 869–888.

Dörr, W., Zulauf, G., Fiala, J., Franke, W., Vejnar, Z., 2002. Neoproterozoic to Early Cambrian history of an active plate margin in the Teplá-Barrandian unit—a correlation of U–Pb isotopic-dilution-TIMS ages (Bohemia, Czech Republic). Tectonophysics 352, 65–85.

Edel, J.B., Weber, K., 1995. Cadomian terranes, wrench faulting and thrusting in the central Europe Variscides: geophysical and geological evidence. Geologische Rundschau 84, 412–432.

Edel, J.B., 1987. Paleopositions of the western Europe Hercynides during the Late Carboniferous deduced from paleomagnetic data: consequences for "stable Europe". In: Kent, D.V., Krs, M. (Eds.), Laurasian Paleomagnetism and Tectonics. Tectonophysics 139, 31–41.

Fisher, R., 1953. Dispersion on a sphere. Proceedings of the Royal Society A 217, 295–305.

Floyd, P.A., Leveridge, B.E., 1987. Tectonic environment of the Devonian Gramscatho basin, south Cornwall: framework mode and geochemical evidence from turbiditic sandstones. Journal of the Geological Society, London 144, 531–542.

Floyd, P.A., Shail, R., Leveridge, B.E., Franke, W., 1991. Geochemistry and provenance of Rhenohercynian sandstones: implications for tectonic environment discrimination. In: Developments in Sedimentary Provenance Studies. In: Morton, A.C., Todd, S.P., Haughton, P.D.W. (Eds.), Special Publications 57. Geological Society, London, pp. 173–188.

Franke, W., Altherr, R., Haak, V., Oncken, O., 1999. Orogenic processes—quantification and modelling in the Variscides: summary of summaries. In: Geologische Vereinigung 89th Annual Meeting, Freiberg, Saxony. Terra Nostra 99, 7–10.

Haq, B.U., Van Eysinga, F.W.B., 1998. Geological Time Table. Elsevier Sciences Publishers, Amsterdam.

Havlíček, V., 1976. Evolution of Ordovician Brachiopod communities in the Mediterranean Province. In: Basset, M.G. (Ed.), The Ordovician System. University of Wales Press and Natural Museum of Wales, Cardiff, pp. 349–358.

Havlíček, V., 1981. Development of a linear sedimentary depression exemplified by the Prague Basin (Ordovician—Middle Devonian, Barrandian area—central Bohemia). Sborník geologických Věd, Geologie 35, 7–48.

Hladil, J., Melichar, R., Otava, J., Galle, A., Krs, M., Man, O., Pruner, P., Čejchan, P., Orel, P., 1999. The Devonian in the easternmost Variscides, Moravia: a holistic analysis directed towards comprehension of the original context. Abhandlungen der Geologische Bundesanstalt, Wien 54, 27–47.

Jakeš, P., Zoubek, J., Zoubková, J., Franke, W., 1979. Graywackes and metagraywackes of the Teplá-Barrandian Proterozoic area. Sborník geologických Věd, Geologie 33, 83–122.

Jelínek, V., 1966. A high sensitivity spinner magnetometer. Studia Geophysica et Geodaetica 10, 58–78.

Jelínek, V., 1973. Precision AC bridge set for measuring magnetic susceptibility and its anisotropy. Studia Geophysica et Geodaetica 17, 36–48.

Kachlík, V., Patočka, F., 1998. Cambrian/Ordovician intracontinental rifting and Devonian closure of the rifting generated basins in the Bohemian Massif realms. Acta Universitatis Carolinae, Geologica 42, 433–441.

Keppie, D., Dostal, J., 1998. Terrane transfer between eastern Laurentia and northwestern Gondwana: the place of the Teplá-Barrandian, Bohemian Massif. Acta Universitatis Carolinae, Geologica 42, 447–450.

Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of palaeomagnetic data. Geophysical Journal of the Royal Astronomical Society 62, 699–718.

Krs, M., 1968. Rheological aspects of palaeomagnetism? In: Proceedings of the 23rd International Geological Congress 5, pp. 87–96.

Krs, M., 1986. Multicomponent analysis of magnetisation and palaeogeographic reconstruction of Early Phanerozoic rocks of the Barrandian, based on palaeomagnetic data. In: Research of deep geological structure of the Czechoslovak territory, pp. 97–113.

Krs, M., Pruner, P., 1995. Palaeomagnetism and palaeogeography of the Variscan formations of the Bohemian Massif, comparison with other European regions. Journal of the Czech Geological Society 40 (1–2), 3–46.

Krs, M., Krsová, M., Pruner, P., 1992. Palaeomagnetic investigations of Upper Cambrian volcanics, Barrandian, Bohemian Massif. In: Vrána, S., Schulmannová, B. (Eds.), Geological Model of Western Bohemia in Relation to the Deep Borehole KTB in the FRG—Abstracts. Czech Geological Survey Prague. Internal unpublished Report, Geofond Prague, pp. 11–13.

Krs, M., Krsová, M., Pruner, P., 1997. Palaeomagnetism and palaeogeography of the Variscan and pre-Variscan formations of the Bohemian Massif. In: Vrána, S., Štědrá, V. (Eds.), Geological Model of Western Bohemia Related to the KTB Borehole in Germany. Sborník geologických Věd Geologie 47, 162–173.

Krs, M., Pruner, P., Man, O., 2001. Tectonic and paleogeographic interpretation of the paleomagnetism of Variscan and pre-Variscan formations of the Bohemian Massif, with special reference to the Barrandian terrane. Tectonophysics 332, 93–114.

Krs, M., Krsová, M., Pruner, P., Havlíček, V., 1986. Palaeomagnetism, palaeogeography and multi-component analysis of magnetization of Ordovican rocks from the Barrandian area of the Bohemian Massif. Sborník geologických věd, Užitá geofyzika 20, 9–45.

Krs, M., Krsová, M., Pruner, P., Havlíček, V., 1988. Palaeomagnetism, magnetism and palaeogeography of the Middle and Upper Cambrian rocks of the Barrandian area in the Bohemian Massif. Sborník geologických věd, Uzitá geofyzika 22, 9–48.

Krs, M., Krsová, M., Pruner, P., Chvojka, R., Havlíček, R., 1987. Palaeomagnetism, palaeogeography and the multicomponent analysis of Middle and Upper Cambrian rocks of the Barrandian in the Bohemian Massif. In: Kent, D.V., Krs, M. (Eds.), Laurasian Paleomagnetism and Tectonics. Tectonophysics 139, 1–20.

Kříbek, B., Pouba, Z., Skoček, V., Waldhausrová, J., 2000. Neoproterozoic of the Teplá-Barrandian Unit as a part of the Cadomian orogenic belt: A review and correlation aspects. Věstník Českého geologického ústavu 75, 175–196.

Kukal, Z., Jäger, O., 1988. Siliciclastic signal of the variscan orogenesis: the Devonian Srbsko Formation of Central Bohemia. Věstník Ústředního Ústavu geologického 63, 65–71.

Kuncíř, J., Benada, J., Řanda, Z., Vobecký, M., 1980. Multi-element standard for routine instrumental activation analysis of trace elements in rocks and tectites. Journal of Radioanalytical Chemistry 5, 369–378.

Mallard, L.D., Rogers, J.J.W., 1997. Relationship of Avalonian and Cadomian terranes to Grenville and Pan-African events. Journal of Geodynamics 23, 197–224.

Matějka, D., 1981. Geochemical investigation of the Proterozoic rocks in the Svojšín and Neveklov areas. In: Pouba, Z. (Ed.), Correlations of Proterozoic and Palaeozoic Stratiform Deposits. Institute of Geological Sciences of Charles University 4, pp. 62–80.

Mauritsch, H.J., Pruner, P., 1990. Paleomagnetic investigations in the central part of the Bohemian Massif Barrandian. In: Minaříková, D., Lobitzer, H. (Eds.), Thirty Years of Geological Cooperation Between Austria and Czechoslovakia. Geological Survey Prague, pp. 56–64.

McLennan, S.M., 2001. Relationships between the trace element composition of sedimentary rocks and upper continental crust. Geochemistry, Geophysics, Geosystems 2, 1–29.

McLennan, S.M., Hemming, S., McDaniel, D.K., Hanson, G.N., 1993. Geochemical approaches to sedimentation, provenance and tectonics. In: Johnsson, M.J., Basu, A. (Eds.), Processes Controlling the Composition of Clastic Sediments. Geological Society of America, Special Paper 284, pp. 21–40.

Nance, D., Murphy, J.B., 1994. Contrasting basement isotopic signatures and the palinspastic restoration of peripheral orogens: Example from the Neoproterozoic Avalonian–Cadomian belt. Geology 22, 617–620.

Nance, W.B., Taylor, S.R., 1976. Rare earth patterns and crustal evolution. Geochimica et Cosmochimica Acta 40, 1539–1551.

Patočka, F., Galle, A., Vavrdová, M., Vlašímský, P., 1994. Early Paleozoic evolution of the Barrandian Terrane, Bohemian Massif, Czech Republic: paleotectonic implications of sedimentary, fossil and volcanic record. Journal of the Czech Geological Society 39, 82–83.

Patočka, F., Vlašímský, P., Blechová, K., 1993. Geochemistry of Early Paleozoic volcanics of the Barrandian Basin (Bohemian Massif, Czech Republic): implications for paleotectonic reconstructions. Jahrbuch der Geologische Bundesanstalt, Wien 136, 871–894.

Pharaoh, T.C., 1999. Palaeozoic terranes and their lithospheric boundaries within the Trans-European Suture Zone (TESZ): a review. Tectonophysics 314, 17–41.

Příhoda, K., Krs, M., Pešina, B., Bláha, J., 1989. MAVACS—a new system creating a non-magnetic environment for palaeomagnetic studies. In: E. Banda (Ed.), Paleomagnetismo—Palaeomagnetism, Cuadernos de Geologia Iberica 1988–1989, 223–250.

Rename, J., Cita, M.B., Dercourt, J., Bouysse, P., Repetto, F.L., Faure-Muret, A. (Eds.), 2000. International Stratigraphic Chart. UNESCO, IUGS.

Strnad, L., Hladil, J., 2001. Geochemistry and composition of the Middle Devonian Srbsko Formation in Barrandian area, Bohemian Massif: A trench or fore-arc strike-slip basin fill with material from volcanic arc or continental margin? Geolines 13, 111–112.

Tait, J., Bachtadse, V., Soffel, H., 1994a. Silurian paleogeography of Armorica: New palaeomagnetic data from central Bohemia. Journal of Geophysical Research 99 (B2), 2897–2907.

Tait, J., Bachtadse, V., Soffel, H., 1994b. New palaeomagnetic constraints on the position of central Bohemia during Early Ordovician times. Geophysical Journal, International 116, 131–140.

Tait, J., Bachtadse, V., Soffel, H., 1995. Upper Ordovician palaeogeography of the Bohemian Massif: Implications for Armorica. Geophysical Journal, International 122, 211–218.

Tait, J., Bachtadse, V., Soffel, H., 1996. Eastern Variscan fold belt: Paleomagnetic evidence for oroclinal bedding. Geology 24, 871–874.

Taylor, S.R., McLennan, S.M., 1985. The Continental Crust: Its Composition and Evolution. Blackwell Science Publications, Oxford. 383 pp.

Torsvik, T.H., Smethurst, M.A., Van der Voo, R., Trench, A., Abrahamsen, N., Halvorsen, E., 1992. Baltica. A synopsis of Vendian–Permian palaeomagnetic data and their palaeotectonic implications. Earth Science Review 33, 133–152.

Torsvik, T.H., Smethurst, M.A., Briden, J.C., Sturt, B.A., 1990. A review of Palaeozoic palaeomagnetic data from Europe and their palaeogeographical implications. In: McKerrow, W.S., Scotese, C.R. (Eds.), Palaeozoic Palaeogeography and Biogeography. Geological Society Memoirs 12, 25–41.

Van der Voo, R., 1990. Phanerozoic palaeomagnetic poles from Europe and North America and comparisons with continental reconstructions. Reviews of Geophysics 28, 167–206.

Van der Voo, R., 1993. Paleomagnetism of the Atlantic, Tethys and Iapetus Oceans. Cambridge Press, Cambridge. 273 pp.

Vidal, Ph., Auvray, B., Charlot, R., Fediuk, F., Hameurt, J., Waldhausrová, J., 1975. Radiometric age of volcanics of the Cambrian "Křivoklát–Rokycany" complex (Bohemian Massif). Geologische Rundschau 64, 563–570.

Vobecký, M., Frána, J., Bauer, J., Řanda, Z., Benada, J., Kuncíř, J., 1971. Radioanalytical determination of elemental composition of lunar samples. In: Proceedings of the Second Lunar Science Conference, vol. 2. The Massachusetts Institute of Technology Press, pp. 1291–1300.

Waldhausrová, J., 1971. The chemistry of the Cambrian volcanics in the Barrandian area. Krystalinikum 8, 45–75.

Winchester, J.A., the PACE TMR Network Team (contract ERB-FMRXCT97-0136) including Belka, Z., Patočka, F., Kachlík, V., 2002. A new interpretation of the Palaeozoic amalgamation of the Central Europe, based on new geological and geophysical investigations. Journal of Geological Society, London, Special Papers, in press.

Ziegler, P.A., 1989. Evolution of Laurussia. A study in Late Paleozoic Plate Tectonics. Kluwer Academic Publishers, Amsterdam, p. 102.

Zulauf, G., Schritter, F., Riegler, G., Finger, F., Fiala, J., Vejnar, Z., 1999. Age constraints on the Cadomian evolution of the Teplá-Barrandian unit (Bohemian Massif) through electron microprobe dating of metamorphic monazite. Zeitschrif der deutschen Geologischen Gesellschaft 180, 627–639.